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- Hierarchical models
 - Rats example
 - Factory example
 - 8 schools example
- Exchangeability
- Computational aspects

Bayesian Statistics and Data Analysis

Lecture 7

Måns Magnusson

Department of Statistics, Uppsala University
Thanks to Aki Vehtari, Aalto University



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- **Hierarchical models**
 - Rats example
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Section 1

Hierarchical models



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Stan tutorial for hierarchical models

- Hierarchical models

- Rats example
- Factory example
- 8 schools example

- Exchangeability

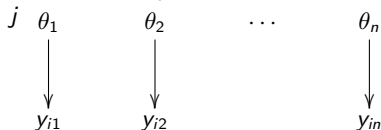
- Computational aspects

- Before Assignment 7, watch [Hierarchical Modelling in Stan: Predicting the Premier League](#)
- While watching, connect the example to:
 - partial pooling
 - group-level parameters
 - comparison with non-hierarchical models
 - Stan diagnostics for hierarchical models



Hierarchical model

- Example: Treatment effectiveness
 - in hospital j the survival probability is θ_j
 - observations y_{ij} tell whether patient i survived in hospital



- **Hierarchical models**
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- Hierarchical models

- Rats example
- Factory example
- 8 schools example

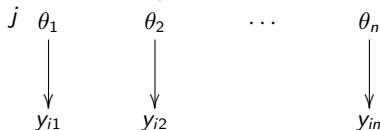
- Exchangeability

- Computational aspects

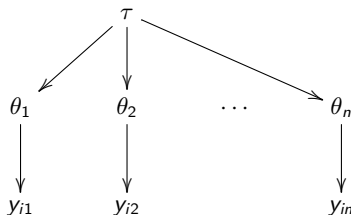
Hierarchical model

- Example: Treatment effectiveness

- in hospital j the survival probability is θ_j
- observations y_{ij} tell whether patient i survived in hospital



- sensible to assume that θ_j are similar

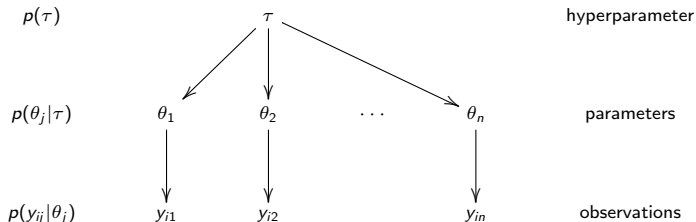


- natural to think that θ_j have common population distribution
- θ_j is not directly observed and the population distribution is unknown



Hierarchical model: terms

Lvl 1: observations given parameters $p(y_{ij}|\theta_j)$



Joint posterior

$$\begin{aligned} p(\theta, \tau|y) &\propto p(y|\theta, \tau)p(\theta, \tau) \\ &\propto p(y|\theta)p(\theta|\tau)p(\tau) \end{aligned}$$



● Hierarchical models

- Rats example
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- 8 schools example

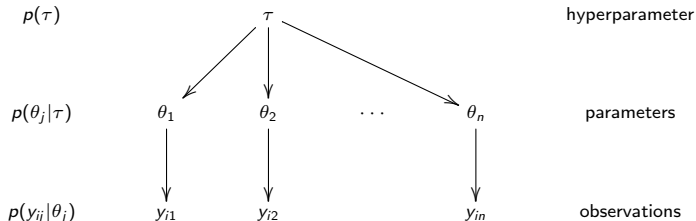
● Exchangeability

● Computational aspects

Hierarchical model: terms

Lvl 1: observations given parameters $p(y_{ij}|\theta_j)$

Lvl 2: parameters given hyperparameters $p(\theta_j|\tau)$



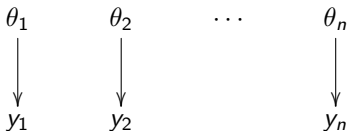
Joint posterior

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Comparisons

- "Separate model" (model with separate/independent effects)



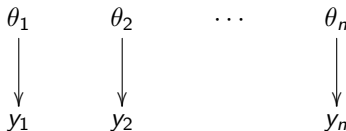
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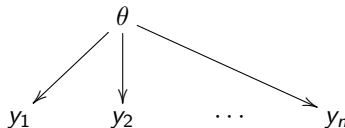
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Comparisons

- "Separate model" (model with separate/independent effects)



- "Joint/pooled model" (model with a common effect / pooled model)





- **Hierarchical models**

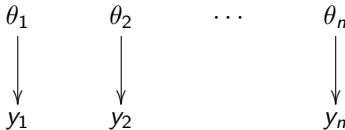
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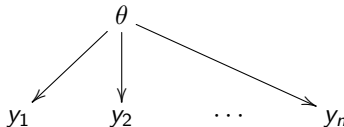
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Comparisons

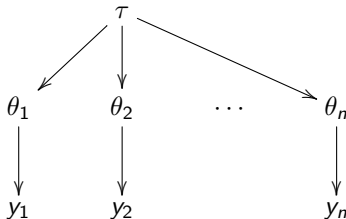
- "Separate model" (model with separate/independent effects)



- "Joint/pooled model" (model with a common effect / pooled model)



- Hierarchical model





Predictive distributions for hierarchical models

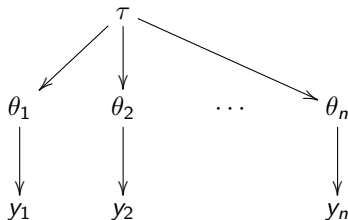
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- Two types of predictive distributions
 1. A new observation in **an existing group**
 2. A new observation in **a new group**





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Hierarchical binomial model: rats

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- Medicine testing
 - Type F344 female rats in control group given placebo
 - count how many get endometrial stromal polyps
 - familiar binomial model example



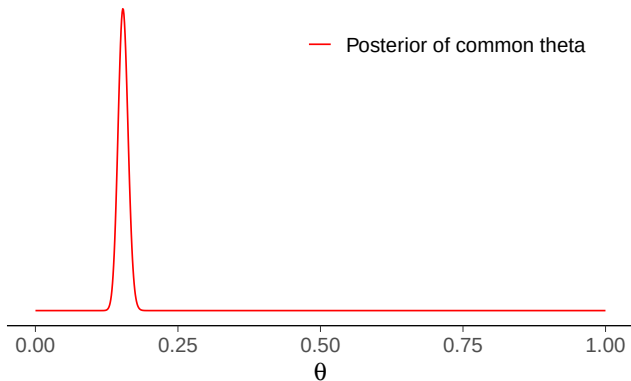
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- Medicine testing
- Type F344 female rats in control group given placebo
 - count how many get endometrial stromal polyps
 - familiar binomial model example
- Experiment has been repeated 71 times

0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/19	0/19	0/19
0/19	0/18	0/18	0/17	1/20	1/20	1/20	1/20	1/19	1/19
1/18	1/18	2/25	2/24	2/23	2/20	2/20	2/20	2/20	2/20
2/20	1/10	5/49	2/19	5/46	3/27	2/17	7/49	7/47	3/20
3/20	2/13	9/48	10/50	4/20	4/20	4/20	4/20	4/20	4/20
4/20	10/48	4/19	4/19	4/19	5/22	11/46	12/49	5/20	5/20
6/23	5/19	6/22	6/20	6/20	6/20	16/52	15/46	15/47	9/24
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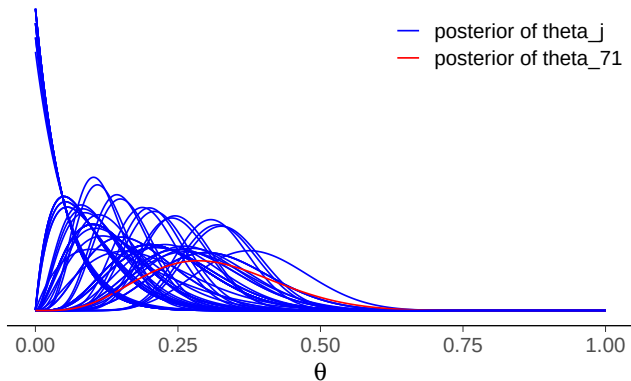
Pooled model



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Separate model



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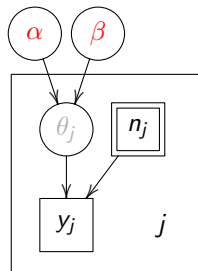


Hierarchical binomial model: rats

- Hierarchical binomial model for rats
prior parameters α and β are unknown

$$\theta_j | \alpha, \beta \sim \text{Beta}(\theta_j | \alpha, \beta)$$

$$y_j | n_j, \theta_j \sim \text{Bin}(y_j | n_j, \theta_j)$$



- Joint posterior $p(\theta_1, \dots, \theta_J, \alpha, \beta | y)$
 - multiple parameters

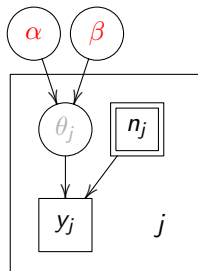


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- Joint posterior $p(\theta_1, \dots, \theta_J, \alpha, \beta | y)$
 - multiple parameters
 - factorize $\prod_{j=1}^J p(\theta_j | \alpha, \beta, y) p(\alpha, \beta | y)$



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- Population prior $\text{Beta}(\theta_j | \alpha, \beta)$
- Hyperprior $p(\alpha, \beta)$?
 - α, β both affect the location and scale
 - BDA3 (p. 110) has (vague) $p(\alpha, \beta) \propto (\alpha + \beta)^{-5/2}$



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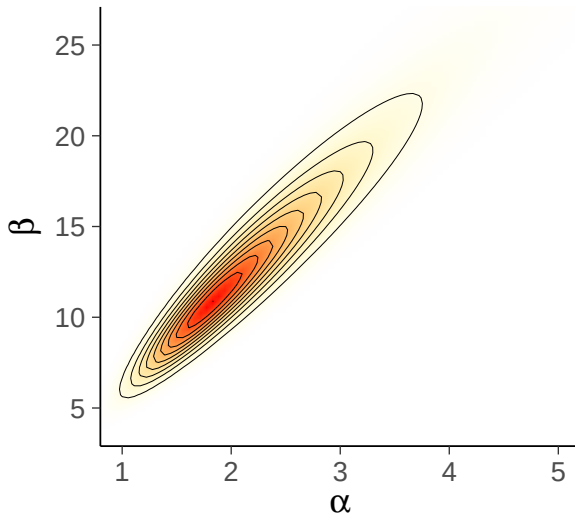
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- What type of predictive distributions can we have?



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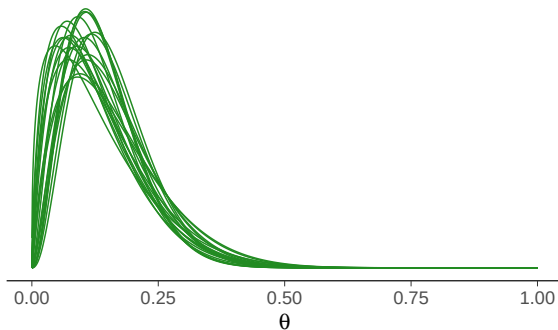
Hierarchical binomial model: rats

The marginal of α and β





Beta(α, β) given posterior draws of α and β



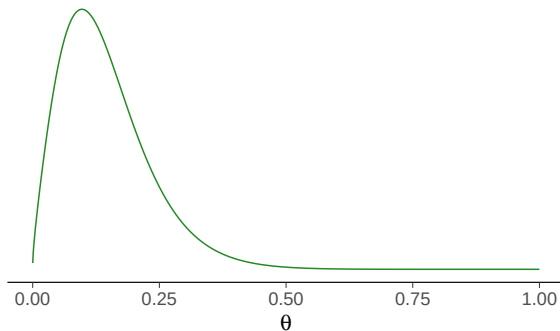
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Hierarchical binomial model: rats

Population distribution (prior) for θ_j



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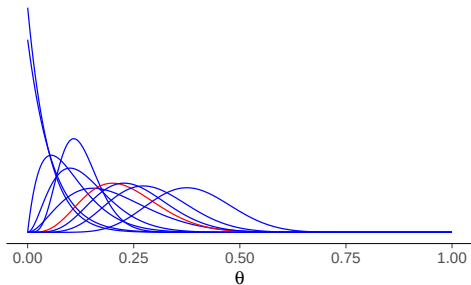


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Hierarchical binomial model: rats

Separate model



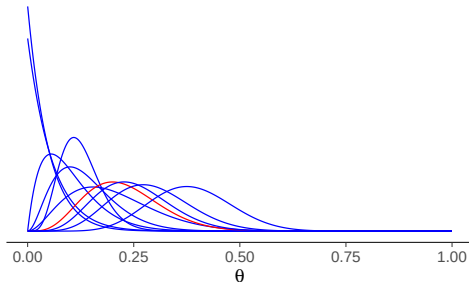


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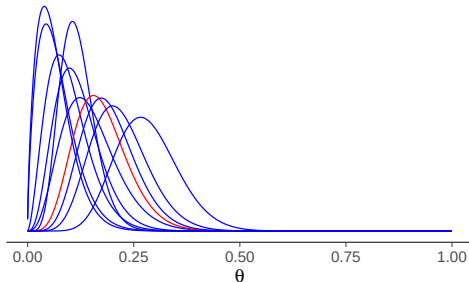
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Hierarchical binomial model: rats

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Hierarchical model



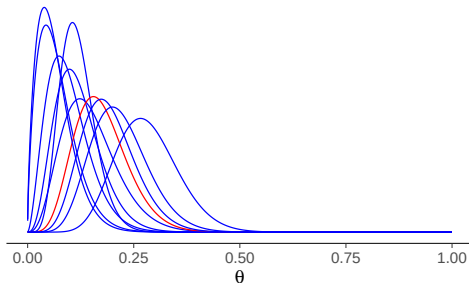


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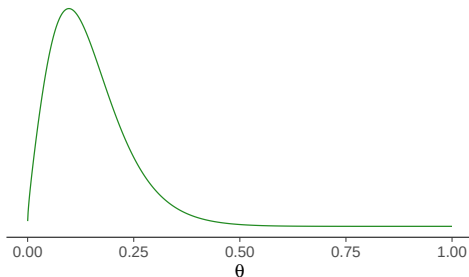
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Hierarchical binomial model: rats

Hierarchical model



Population distribution (prior) for θ_j





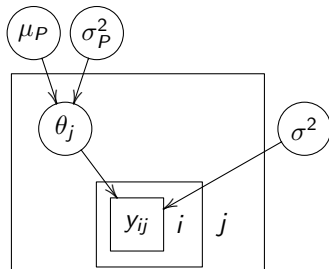
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Hierarchical normal model: factory

- Factory has 6 machines which quality is evaluated
- Assume hierarchical model
 - each machine has its own (average) quality θ_j and **common variance** σ^2

$$\theta_j | \mu_P, \sigma_P^2 \sim \mathcal{N}(\mu_P, \sigma_P^2)$$

$$y_{ij} | \theta_j \sim \mathcal{N}(\theta_j, \sigma^2)$$



- Can be used to predict the future quality produced by each machine and quality produced by a new similar machine



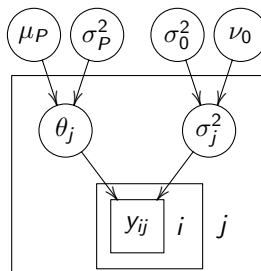
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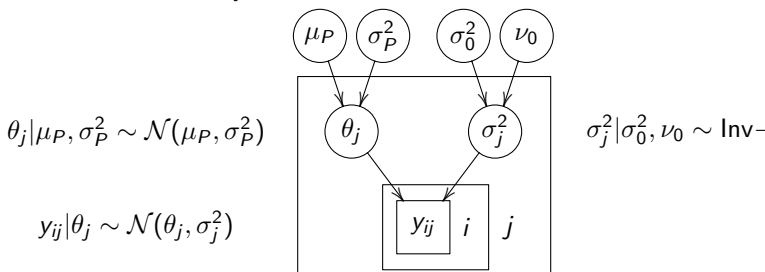
$$\sigma_j^2 | \sigma_0^2, \nu_0 \sim \text{Inv-}$$



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Hierarchical normal model: factory

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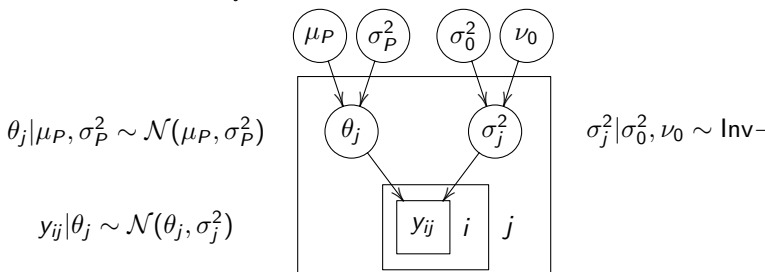
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Hierarchical normal model: factory

- Factory has 6 machines which quality is evaluated
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- **What type of predictive distributions can we have?**
- Can be used to predict the future quality produced by each machine and quality produced by a new similar machine



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- Example: SAT coaching effectiveness
 - in USA commonly used Scholastic Aptitude Test (SAT) is designed so that short term practice should not improve the results significantly
 - schools have anyway coaching courses
 - test the effectiveness of the coaching courses



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- Example: SAT coaching effectiveness
 - in USA commonly used Scholastic Aptitude Test (SAT) is designed so that short term practice should not improve the results significantly
 - schools have anyway coaching courses
 - test the effectiveness of the coaching courses
- SAT
 - standardized multiple choice test
 - mean about 500 and standard deviation about 100
 - most scores between 200 and 800
 - different topics, e.g., V=Verbal, M=Mathematics
 - pre-test PSAT



- Effectiveness of the SAT coaching
 - students had made pre-tests PSAT-M and PSAT-V
 - part of students were coached
 - linear regression was used to estimate the coaching effect y_j for the school j (could be denoted with $\bar{y}_{.j}$, too) and variances σ_j^2
 - y_j approximately normally distributed, with variances assumed to be known based on about 30 students per school
 - data is group means and variances (not personal results)

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• Data:	School	A	B	C	D	E	F	G	H
	y_j	28	8	-3	7	-1	1	18	12
	σ_j	15	10	16	11	9	22	20	28



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Hierarchical normal model for group means

- J experiments, unknown θ_j and known σ^2

$$y_{ij}|\theta_j \sim \mathcal{N}(\theta_j, \sigma^2), \quad i = 1, \dots, n_j; \quad j = 1, \dots, J$$

- Group j sample mean and sample variance

$$\bar{y}_{.j} = \frac{1}{n_j} \sum_{i=1}^{n_j} y_{ij}$$

$$\sigma_j^2 = \frac{\sigma^2}{n_j}$$



Hierarchical normal model for group means

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$$\sigma_j^2 = \frac{\sigma^2}{n_j}$$

- Use model

$$\bar{y}_{.j}|\theta_j \sim \mathcal{N}(\theta_j, \sigma_j^2)$$

this model can be generalized so that, σ_j^2 can be different from each other for other reasons than n_j

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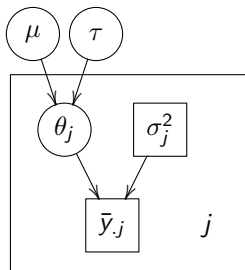


Hierarchical normal model for group means

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$$\theta_j | \mu, \tau \sim \mathcal{N}(\mu, \tau)$$

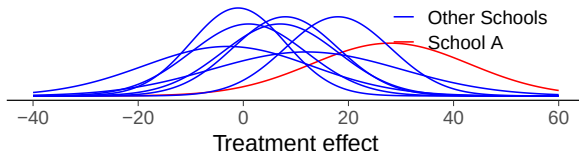
$$\bar{y}_j | \theta_j \sim \mathcal{N}(\theta_j, \sigma_j^2)$$





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Separate model

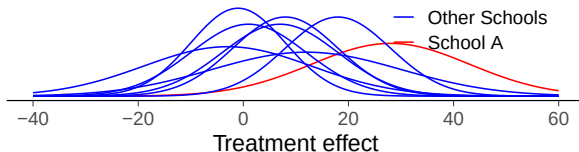




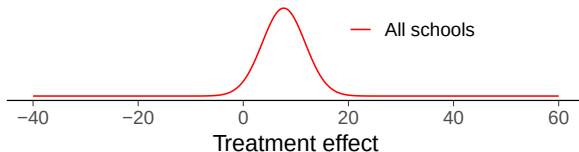
Hierarchical normal model: 8 schools

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Separate model



Pooled model





- Hierarchical models

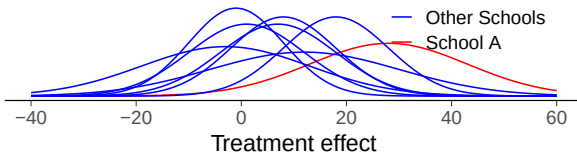
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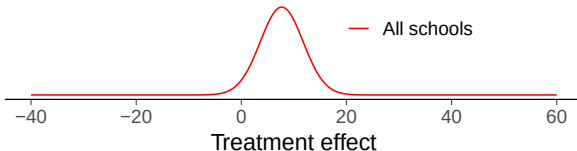
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Hierarchical normal model: 8 schools

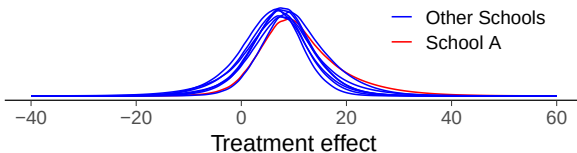
Separate model



Pooled model



Hierarchical model

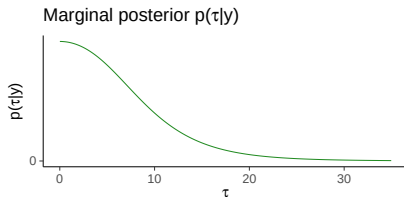




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Hierarchical normal model: 8 schools

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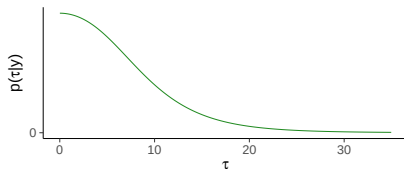




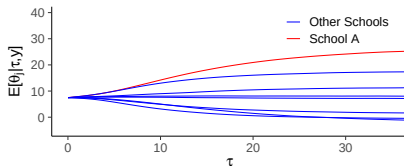
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Hierarchical normal model: 8 schools

Marginal posterior $p(\tau|y)$



Conditional means $E[\theta_i|\tau, y]$



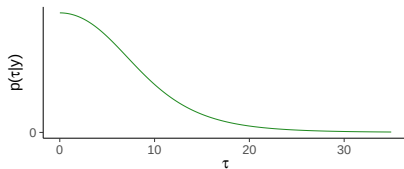


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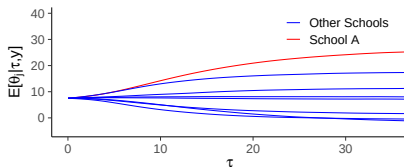
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Hierarchical normal model: 8 schools

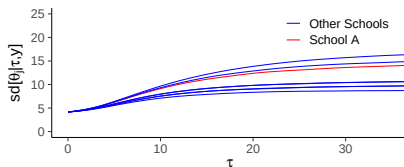
Marginal posterior $p(\tau|y)$



Conditional means $E[\theta_j|\tau, y]$



Conditional standard deviations $sd[\theta_j|\tau, y]$





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Section 2

Exchangeability



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Exchangeability

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- Justifies why we can use
 - a joint model for data
 - a joint prior for a set of parameters
- Less strict than independence (IID)
- IID implies exchangeability



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Exchangeability

- *Exchangeability*

Random variables $\theta_1, \dots, \theta_J$ (or $y_1, \dots, y_J$) are **exchangeable** if the joint distribution p is invariant to the permutation of indices

i.e

$$p(\theta_1, \theta_2, \theta_3) = p(\theta_{\pi(1)}, \theta_{\pi(2)}, \theta_{\pi(3)})$$

for any permutation π of the indices. E.g.

$$p(\theta_1, \theta_2, \theta_3) = p(\theta_2, \theta_3, \theta_1)$$



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 - 8 schools example
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Exchangeability

- *Exchangeability*

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E.g. when we have a trend over indices (like in time series data)

- Exchangeability implies symmetry:
If there is no information which can be used *a priori* to separate θ_j from each other, we can assume exchangeability. ("Ignorance implies exchangeability")



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Exchangeability: DeFinettis theorem

Let $(x_n)_{n=1}^{\infty}$ to be an infinite sequence of exchangeable random variables. De Finetti's theorem then says that there is some random variable θ so that x_j are conditionally independent given θ , and joint density for $x_1, \dots, x_J$ can be written in the *iid mixture* form

$$p(x_1, \dots, x_J) = \int \left[\prod_{j=1}^J p(x_j | \theta) \right] p(\theta) d\theta$$

- **Exchangeability**: the data behave as if they were i.i.d., once we account for some unknown parameter θ



Example

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- Suppose we're observing coin flips, but we're not sure about the bias θ
 - We think the flips are 'the same kind of thing' (no reason to treat flip 1 differently from flip 2): **Exchangeability**
 - But we might not believe they are strictly independent because we're uncertain about the bias: they share information through the unknown θ



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- Suppose we're observing coin flips, but we're not sure about the bias θ
 - We think the flips are 'the same kind of thing' (no reason to treat flip 1 differently from flip 2): **Exchangeability**
 - But we might not believe they are strictly independent because we're uncertain about the bias: they share information through the unknown θ
- Given θ the flips are i.i.d.
- Without conditioning on θ , the flips are only exchangeable, not iid.



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- Exchangeability does not mean that the results of the experiments could not be different
 - e.g. if we know that the experiments have been in two different laboratories, and we know that the other laboratory has better conditions for the rats, but we do not know which experiments have been made in which laboratory
 - a priori experiments are exchangeable
 - model could have unknown parameter for the laboratory with a conditional prior for rats assumed to come from the same place (clustering model)



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Exchangeability and additional information

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- Example: bioassay
 - y_i number of dead animals are not exchangeable alone



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 - x_i dose is additional information



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- Example: bioassay
 - y_i number of dead animals are not exchangeable alone
 - x_i dose is additional information
 - (x_i, y_i) exchangeable and logistic regression was used

$$p(\alpha, \beta | y, n, x) \propto \prod_{i=1}^n p(y_i | \alpha, \beta, n_i, x_i) p(\alpha, \beta)$$



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Hierarchical exchangeability

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 - Example: hierarchical rats example
 - all rats not exchangeable



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Hierarchical exchangeability

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 - all rats not exchangeable
 - in a single laboratory rats exchangeable



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Hierarchical exchangeability

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- Example: hierarchical rats example
 - all rats not exchangeable
 - in a single laboratory rats exchangeable
 - laboratories exchangeable
 - hierarchical model can be used



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- Conditional exchangeability
 - if y_i is connected to an additional information x_i , so that y_i are not exchangeable, but (y_i, x_i) exchangeable use joint model or conditional model $(y_i|x_i)$.



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- Conditional exchangeability
 - if y_i is connected to an additional information x_i , so that y_i are not exchangeable, but (y_i, x_i) exchangeable use joint model or conditional model $(y_i|x_i)$.
- Partial exchangeability
 - if the observations can be grouped (a priori), then we can use a hierarchical model



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Section 3

Computational aspects



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- Hierarchical models

- Group-level or global parameters, e.g.

$$\tau \sim p(\tau)$$

- Local or individual-level parameters

$$\theta_i \sim \mathcal{N}(0, \tau)$$

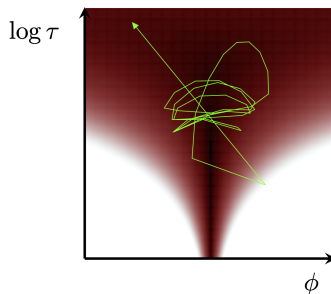
- Creates a "funnel-like" posterior geometry:
- Comes from the variance in the different layers:
 - When τ is small, the θ_i 's are concentrated around 0
 - When τ is large, the θ_i 's are widely dispersed



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Why problematic?

1. Pathological geometry: difficult to explore efficiently
2. Divergences: HMC will risk divergencies (also a good diagnostic)



Betancourt (2020)

demo



Handling the funnel

1. Reduce step size (adapt_delta closer to 1)
2. Reparametrize using non-centered parametrization

2.1 Centered parametrization

$$\theta_i \sim N(\mu, \tau)$$

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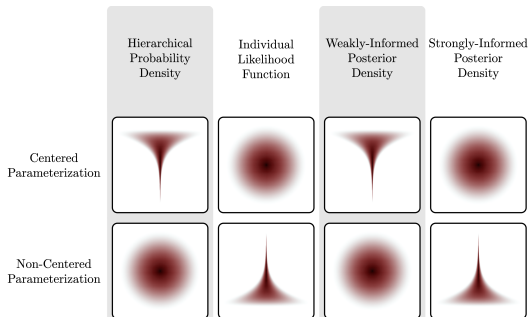
2.1 Centered parametrization

$$\theta_i \sim N(\mu, \tau)$$

2.2 Non-centered parametrization

$$\eta_i \sim N(0, 1)$$

$$\theta_i = \mu + \tau \eta_i$$



Betancourt (2020)